

A positive characteristic root disproves conjecture 00000008535

Local AI-assisted solution draft
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Claim being disproved. The first clause asserts that every root of the characteristic polynomial of a supersolvable geometric lattice is real and negative. With the standard characteristic-polynomial convention, the two-element Boolean lattice already disproves negativity.

The lattice and its hypotheses. Let $L = \{0, 1\}$ with $0 < 1$. Its meet and join are minimum and maximum; its rank function is $r(0) = 0$, $r(1) = 1$. It is atomistic: 1 is its only atom, 0 is the empty join of atoms, and 1 is the join of itself. It is upper semimodular. Indeed, if x covers $x \wedge y$, then necessarily $x = 1$ and $y = 0$, and $x \vee y = 1$ covers $y = 0$. Thus L is a finite geometric lattice.

Both elements are modular: for $a \leq b$, the identity $a \vee (m \wedge b) = (a \vee m) \wedge b$ holds for $m = 0$ and $m = 1$. Therefore the maximal chain $0 < 1$ consists entirely of modular elements, and L is supersolvable.

Characteristic polynomial. The defining recurrence

$$\sum_{y \leq x} \mu(0, y) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0 \end{cases}$$

gives $\mu(0, 0) = 1$ and $\mu(0, 1) = -1$. Hence

$$\chi_L(t) = \sum_{x \in L} \mu(0, x) t^{r(1) - r(x)} = t - 1.$$

It has the positive real root $t = 1$, contrary to the asserted strict negativity. This suffices to disprove the whole conjecture, regardless of its additional root-modulus assertion.

Formal verification and semantic correspondence. The accompanying `Main.lean` is self-contained over Lean 4.19.0 Std. Its finite `LatticeData` contains an order, bottom, top, meet, join, and rank. It checks the partial-order and lattice universal properties, the grading of covers, atomisticity, and upper semimodularity. Supersolvability is expressed by a maximal chain all of whose elements satisfy the modular-element identity. These properties are proved for `booleanOne`.

The file checks the defining Möbius recurrence for the actual two values and evaluates the characteristic polynomial by its defining sum. The theorem `positive_characteristic_root` proves that this sum vanishes at the integer 1. The universal proposition `NegativeRootClaim` is the necessary integer-root specialization of the conjecture's first clause: every integer root of every supersolvable geometric lattice is negative. Every integer root is a real root under the canonical inclusion $\mathbb{Z} \subset \mathbb{R}$; therefore falsity of this specialization refutes the original real-root statement. No analytic assertion about real roots is needed.

The final theorem `conjecture8535_false` negates `NegativeRootClaim` using all the verified hypotheses and the root 1. Kernel-checked finite enumeration uses `decide`. The printed axiom lists contain only the standard axiom `propext`; there are no proof placeholders, external decision oracles, or additional axioms.

Reproduction. Run `lean Main.lean` using Lean 4.19.0.